

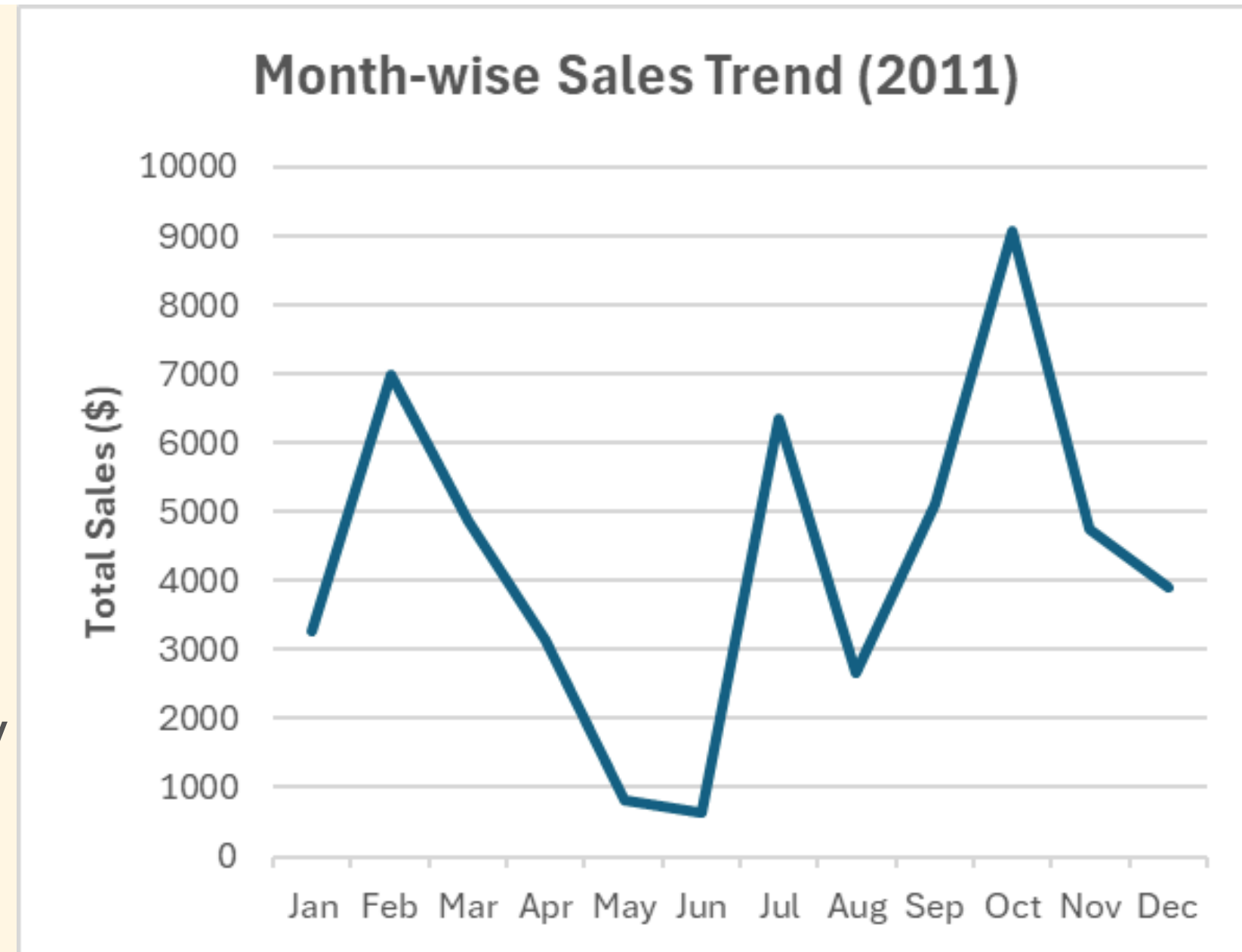
Retail Sales Analysis (SQL Project)

Turning raw retail data into actionable business insights

- Relational database design from raw sales data
 - Business-driven insights using SQL
 - Joins, aggregations & CASE logic

Month-wise Sales Trend (SQL Analysis)

- **Month-wise Sales Trend (SQL Analysis)**
- Peak sales in October (~\$9K)
- Sales show clear seasonality with noticeable peaks in certain months.
- Understanding these trends helps in inventory planning and promotions.
- **Key Insight:**
- Monthly sales are not uniform — timing matters for business decisions.



```
SELECT
    SUBSTRING(order_date, 1,7) AS month,
    ROUND(SUM(CAST(od.sales AS DECIMAL(10,2))),2) AS total_sales
FROM orders o
JOIN order_details od ON o.order_id = od.order_id
GROUP BY SUBSTRING(order_date, 1,7)
ORDER BY month;
```

Repeat vs One-Time Customers (SQL Analysis)

- The majority of customers are one-time buyers.
- Repeat customers account for **~17%** of customers but drive disproportionate long-term value.
-
- Retention-focused strategies can significantly improve revenue stability.
- Insight derived from order-frequency analysis using SQL aggregations.

customer_type	total_customers
One-Time Customers	214
Repeat Customers	43

```
SELECT
  CASE
    WHEN order_count = 1 THEN 'One-Time Customers'
    ELSE 'Repeat Customers'
  END AS customer_type,
  COUNT(*) AS total_customers
FROM (
  SELECT
    customer_name,
    COUNT(DISTINCT order_id) AS order_count
  FROM orders
  GROUP BY customer_name
) t
GROUP BY customer_type;
```

Top Revenue Customers (SQL Analysis)

- **Insight:**
- Revenue is highly concentrated — a small group of customers drives a large share of total sales.
- **Key Finding:**
- Top 10 customers contribute **~17%** of total revenue.
- Highest contributor: Ken Lonsdale (**\$2.3K**).
- **Business Implication:**
- Targeted retention and loyalty programs for top customers can reduce revenue concentration risk.
- **Tech:** SQL joins, aggregations, window functions

customer_name	total_sales
Ken Lonsdale	\$2,377
Maria Etezadi	\$1,803
Paul Stevenson	\$1,768
Erin Smith	\$1,637
Hunter Lopez	\$1,568
Elpida Rittenbach	\$1,559
Dorothy Dickinson	\$1,373
Chris Selesnick	\$1,353
Roy Phan	\$1,348
Christopher Conant	\$1,325

	top_10_revenue_share
▶	17.11

Low-Performing Products (SQL Analysis)

- **Low-Performing Products (SQL Analysis)**

- A small group of products generates negative profit despite steady sales.

-
- These items may need pricing review, promotion, or discontinuation.

- **Business Use:**

- Helps optimize product portfolio and reduce losses.

product_name	total_sales	total_profit
Hewlett Wireless Fax, Laser	1482	-536.23
Smead Lockers, Industrial	1034	-111.48
Ikea Classic Bookcase, Metal	987	-1011.64
Hon Conference Table, Adjustable Height	862	-1059.19
Sharp Wireless Fax, Color	697	-343.1
Bevis Computer Table, Fully Assembled	692	-207.73
Novimex Executive Leather Armchair, Black	668	-219.88
Stanley Canvas, Blue	629	-63.68
SAFCO Executive Leather Armchair, Adjustable	620	-1240.31
Motorola Signal Booster, Cordless	584	-58.56

```
SELECT
    p.product_name,
    ROUND(SUM(CAST(od.sales AS DECIMAL(10,2))),2) AS total_sales,
    ROUND(SUM(CAST(od.profit AS DECIMAL(10,2))),2) AS total_profit
FROM products p
JOIN order_details od ON p.product_id = od.product_id
GROUP BY p.product_name
HAVING total_profit < 0
ORDER BY total_sales DESC
LIMIT 10;
```